Vesugen

Speculative: 200 mcg subcutaneously daily for 10–20 days

Vesugen (KED) is a synthetic tripeptide from the Khavinson bioregulator research program. Plausible applications include peripheral perfusion, vascular erectile function and selected neuronal structural protection. Small human reports describe peripheral and penile blood-flow changes; mouse and human-cell experiments support specific neuronal structural effects. [4–9] Proposed actions involve cellular gene regulation, including endothelial endothelin-1, connexin and SIRT1 expression, rather than a demonstrated classical receptor-agonist mechanism. [1–3] Increased proliferation markers in cell cultures support a theoretical cancer-related precaution. [2]

Human safety is insufficiently characterized, especially for long-term exposure and repeated injected courses. [5–7,10]

⭐ Plausible vascular benefits and selected neuronal structural protection, with human, animal and cell evidence differing by endpoint. [1,4–6,8,9]

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

Available small human reports provide limited treatment experience. [5–7]

Endothelial findings are principally laboratory observations. Small clinical reports support plausible vascular benefits, while neuronal structural protection has animal and cell support. [1,2,4–9]

Evidence Scope

For Vesugen, most human claims originate from small regional reports, occupational studies, peptide combinations, or later reviews rather than accessible randomized trials of Vesugen alone. Those findings are treated as evidence, but they do not receive the same weight as well-controlled and independently replicated human research. [5–7,14]

Dosage & Patterns

The commercial oral formulation and mouse intraperitoneal experiments represent different exposures. The HED below is a body-surface-area comparison, not a validated subcutaneous regimen. [4,10,11]

The mouse Alzheimer's disease study (2021) study used HED **2.21 mg/day**. The rat hypobaric ypoxia study (2008) used **HED 110 mcg/day**. The rat radiation injury study (2010) used **HED 38 mcg/day**.

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Informed by this range of texted animal doses, a common community dose and pattern is

- **Vesugen: 200 mcg subcutaneously daily for 10–20 days, with possible annual repetition.**

Higher doses are also discussed. In the absence of PK and human evidence, speculative experimental higher-doses are also discussed

- **Vesugen high dose: 0.5–2 mg subcutaneously.**

The typical pattern most discussed is daily for 10–20 days, conceived as a brief biological pulse with possible annual repetition. But longer patterns are also discussed, like daily for 4–12 or more weeks.

Pattern Notes

Speculative combination pattern: overlapping or sequential courses with Epitalon, Prostamax, Pinealon or other short-peptide bioregulators. [14]

Benefits

- **Plausible endothelial support.** Vesugen altered endothelin-1, connexin and SIRT1 expression in cultured vascular endothelium, supporting a vascular-regulation hypothesis. [1]
- **Plausible preservation of selected neuronal structures.** In the mouse Alzheimer model, KED improved the pooled proportion of mushroom spines, but the pooled increase in total spine density was not significant. Males showed improved density and mushroom-spine measures; females showed a change in thin spines without significant density or mushroom-spine improvement. Electrophysiological recovery was a nonsignificant trend. Duplicated male/female images were replaced in a 2025 correction; the authors reported unchanged conclusions. [4,8]
- **Plausible support for neuronal structure.** In induced neurons derived from older human donors' fibroblasts, KED increased primary processes by 32% and total dendritic length by 42%; the increase in branch points was not statistically significant. Mitochondrial and lysosomal activity and the measured cellular-aging markers were not significantly changed. These are human-cell findings. [9] Neuronal findings provide a biological rationale for plausible cognitive and stress-adaptation support, while small geriatric and occupational reports describe functional or psychoemotional changes with Vesugen-containing approaches. Combination treatment limits attribution. [4,7,9,14]
- **Plausible improvement in peripheral perfusion and vascular erectile function.** Two small reports, each describing 41 patients, recorded favorable clinical and ultrasound blood-flow changes in previously operated lower-limb ischemia and vasculogenic erectile dysfunction. The indexed title of the erectile-function report refers to lower limbs, although its abstract describes penile-flow outcomes. [5,6]

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Investigational Uses

- **Endothelial regulation and vascular aging** are research targets supported by cell experiments. [1,2]
- **Peripheral arterial insufficiency, vascular erectile dysfunction, and neuronal resilience** are investigational applications, with evidence ranging from small clinical reports to animal and cell studies. [4–7,9]

Cautions

- **Cancer or antiangiogenic therapy:** Higher Ki-67 in endothelial cultures raises a mechanism-based proliferation precaution. Its relevance to occult tumors or cancer treatment is uncertain; it is not evidence that Vesugen causes cancer. [2]
- **Acute ischemic symptoms:** New chest pain or rapidly worsening limb symptoms require urgent clinical assessment. The small chronic vascular reports do not address acute ischemia treatment. [5,6]
- **Concurrent treatment:** Interactions with vasodilators, blood-pressure drugs, antiplatelets, anticoagulants and other peptides are insufficiently characterized. Medication review is appropriate; the cited reports do not establish a specific bleeding interaction. [5–7]
- **Pregnancy and breastfeeding:** Reproductive and infant safety are unresolved; the available human reports address other populations. [5–7]

Biological Mechanism

Vesugen consists of lysine, glutamic acid and aspartic acid. In endothelial cultures, KED altered endothelin-1, connexins and SIRT1; another study found increased Ki-67 and used molecular docking to propose interaction with the MKI67 promoter. Docking supports a binding hypothesis rather than demonstrating target engagement in a treated person. [1,2] KED increased nestin and GAP43 in human periodontal-ligament stem-cell cultures, supporting a neuronal-differentiation hypothesis. [12] Human mesenchymal stem-cell experiments showed context-dependent gene responses, including reduced FOXO1 expression, opposite TNKS2 responses in two aging models and increased NFκB expression. [13] Redox effects are also context-dependent: an early in vitro study found no direct antioxidant activity and observed increased intracellular reactive oxygen species alongside reduced neuronal cell death. [15] These findings favor a model of cellular regulation rather than uniform antioxidant or anti-inflammatory action. Human PK and the link between these laboratory changes and functional benefit remain unresolved. [1–4,12,13,15]

Conclusions

Vesugen has speculative vascular and healthspan-related applications, supported by endothelial experiments and small human reports. Selected neuronal structural effects are

supported by a mouse model and human-cell experiments, with substantial endpoint and sex dependence. The dosing proposals remain speculative. [1–11]

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